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WEEK 2

Module 5 - Python Backend Frameworks – Django, Flask & Fast API

HANDS-ON 1 Web Framework Foundations & Django
Project Setup

Task 1: Understand the Request-Response Cycle Goal: Map out how a web framework handles an HTTP request from browser to response

I did command in command prompt

1. Request-Response Cycle (GET /api/courses/)

Browser sends GET request



URL Router (urls.py)



View (views.py)



Model (models.py) queries the database



View processes the data



Response (HTML/JSON) is returned to the browser

2. Middleware

Middleware sits between the request and the view.

It processes requests before they reach the view and processes responses before they are sent back to the client.

Two built-in Django middleware classes:

(a) SecurityMiddleware

- Adds security features such as HTTPS redirects and security headers.

(b) SessionMiddleware

- Manages user sessions and stores session data.

3. WSGI vs ASGI

WSGI (Web Server Gateway Interface)

- Handles synchronous applications.
- Processes one request at a time.
- Django uses WSGI by default.

ASGI (Asynchronous Server Gateway Interface)

- Supports asynchronous programming.
- Handles WebSockets, long-lived connections, and async requests.
- Use ASGI when building real-time applications like chat apps or live notifications.

4. MVC vs MVT

MVC (Model-View-Controller)

Model -> Database

View -> User Interface

Controller -> Business Logic

Django uses MVT (Model-View-Template)

Model -> Database

View -> Business Logic (acts like Controller)

Template -> User Interface (acts like View in MVC)

MY WORK FLOW

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>python --version
Python 3.14.6

C:\Users\HP>pip install django
'pip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\HP>python -m pip --version
pip 26.1.2 from C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pip (python 3.14)

C:\Users\HP>python -m pip install django
Collecting django
  Downloading django-6.0.6-py3-none-any.whl.metadata (3.9 kB)
Collecting asgiref>=3.9.1 (from django)
  Downloading asgiref-3.11.1-py3-none-any.whl.metadata (9.3 kB)
Collecting sqlparse>=0.5.0 (from django)
  Downloading sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
Collecting tzdata (from django)
  Downloading tzdata-2026.2-py2.py3-none-any.whl.metadata (1.4 kB)
Downloading django-6.0.6-py3-none-any.whl (8.4 MB)
8.4/8.4 MB 2.1 MB/s 0:00:04
Downloading asgiref-3.11.1-py3-none-any.whl (24 kB)
Downloading sqlparse-0.5.5-py3-none-any.whl (46 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
0/4 [tzdata] WARNING: The script sqlformat.exe is installed in 'C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
3/4 [django] WARNING: The script django-admin.exe is installed in 'C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed asgiref-3.11.1 django-6.0.6 sqlparse-0.5.5 tzdata-2026.2

C:\Users\HP>python -m django startproject coursemanager

C:\Users\HP>cd coursemanager

C:\Users\HP\coursemanager>python manage.py runserver

C:\Users\HP>cd coursemanager

C:\Users\HP\coursemanager>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
June 26, 2026 - 15:12:28
Django version 6.0.6, using settings 'coursemanager.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/6.0/howto/deployment/

C:\Users\HP\coursemanager>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
```

```

Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK

C:\Users\HP\coursemanager>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
June 26, 2026 - 15:14:42
Django version 6.0.6, using settings 'coursemanager.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/6.0/howto/deployment/
[26/Jun/2026 15:15:53] "GET / HTTP/1.1" 200 12068
Not Found: /favicon.ico
[26/Jun/2026 15:15:53] "GET /favicon.ico HTTP/1.1" 404 2215

```

MY Django project



The install worked successfully! Congratulations!

View [release notes](#) for Django 6.0

You are seeing this page because `DEBUG=True` is in your settings file and you have not configured any URLs.

django

Task 2: Scaffold and Explore the Django Project

Goal: Create a Django project and a reusable app, and understand the generated file structure.

Step 5: Create Django Project

Command Used

```
django-admin startproject coursemanager
```

One-line comments about generated files

settings.py

Stores all project settings and configuration.

urls.py

Defines URL routes for the Django project.

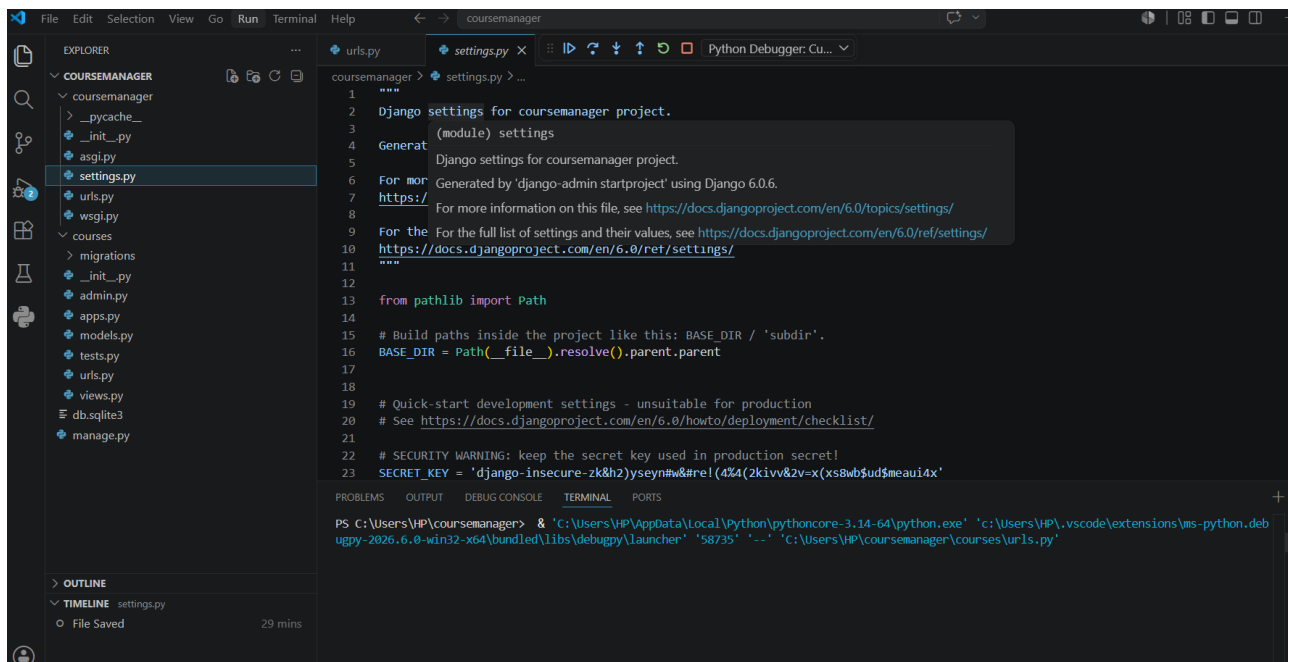
wsgi.py

Entry point for WSGI-compatible web servers.

asgi.py

Entry point for ASGI-compatible web servers.

MY WORKFLOW



Step 6: Create Django App

Command Used

python manage.py startapp courses

Difference between Django Project and Django App

Django Project

- Overall configuration of the application.
- Can contain multiple apps.

Django App

- A reusable module that performs a specific function.
- Contains its own models, views, URLs, and templates.

Step 7: Register the App

In settings.py

```
INSTALLED_APPS = [  
    'courses',  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
]
```

(If you added 'courses' at the end instead of the beginning, that's also perfectly fine.)

Step 8: Create View

File: **courses/views.py**

```
from django.http import HttpResponse  
  
def hello_view(request):  
    return HttpResponse("Course Management API is  
running")
```

Step 9: Create URL Mapping

[courses/urls.py](#)

```
from django.urls import path  
from .views import hello_view
```

```
urlpatterns = [  
    path('hello/', hello_view),  
]  
  
coursemanager/urls.py  
  
from django.contrib import admin  
from django.urls import path, include  
  
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path('api/', include('courses.urls')),  
]
```

Step 10: Run the Server Command

python manage.py runserver

Open

<http://127.0.0.1:8000/api/hello/>

COURSE MANAGER

Urls.py

MY WORKFLOW

```
1 """
2 URL configuration for coursemanager project.
3
4 The 'urlpatterns' list routes URLs to views. For more information please see:
5     https://docs.djangoproject.com/en/6.0/topics/http/urls/
6 Examples:
7 Function views
8     1. Add an import: from my_app import views
9     2. Add a URL to urlpatterns: path('', views.home, name='home')
10 Class-based views
11     1. Add an import: from other_app.views import Home
12     2. Add a URL to urlpatterns: path('', Home.as_view(), name='home')
13 Including another URLconf
14     1. Import the include() function: from django.urls import include, path
15     2. Add a URL to urlpatterns: path('blog/', include('blog.urls'))
16 """
17 from django.contrib import admin
18 from django.urls import path
19
20 urlpatterns = [
21     path('admin/', admin.site.urls),
22 ]
23
```

PS C:\Users\HP\coursemanager> & 'C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58735' '--' 'C:\Users\HP\coursemanager\courses\urls.py'

Veiws.py

```
1 from django.shortcuts import render
2
3 from django.http import HttpResponseRedirect
4
5 def hello_view(request):
6     return HttpResponseRedirect("Course Management API is running")
7
```

PS C:\Users\HP\coursemanager> & 'C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58735' '--' 'C:\Users\HP\coursemanager\courses\urls.py'

OUTPUT

Browser showing:

Course Management API is running